

Section 11.8: Power Series

Recall that the geometric series $\sum_{n=0}^{\infty} ax^n = a + ax + ax^2 + ax^3 + \cdots = \frac{a}{1-x}$ is

convergent when $|x| < 1$. Now we consider the more general situation where the coefficients of x^n are different. That is, we have a series of the form

$$c_0 + c_1x + c_2x^2 + c_3x^3 + \cdots.$$

Then it is not as simple as a geometric series. We can have that it is convergent only when $x = 0$, or when $|x| < R$, or for all x .

More general, a power series centred at a is a series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots,$$

where $a, c_0, c_1, c_2, \cdots$ are constants.

Fact. Every power series is convergent at the centre.

Example 1. Find all values of x such that the power series is convergent.

$$(1) \sum_{n=0}^{\infty} n!x^n.$$

For $x \neq 0$, $\left| \frac{a_{n+1}}{a_n} \right| = (n+1)|x| \rightarrow \infty$. So, the power series is convergent only at $x = 0$.

$$(2) \sum_{n=1}^{\infty} \frac{(x-3)^n}{n}.$$

For $x \neq 3$, $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x-3|}{1 + \frac{1}{n}} \rightarrow |x-3|$. So, the power series is convergent if $|x-3| < 1$

and is divergent if $|x-3| > 1$. If $|x-3| = 1$, then $x = 4$ or $x = 2$.

When $x = 4$, $\sum_{n=1}^{\infty} \frac{1}{n}$ is the divergent harmonic series; when $x = 2$, $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is a

convergent alternating series. Therefore, the power series is convergent on the interval $[2, 4)$.

$$(3) \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n!}.$$

For $x \neq 0$, $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|}{n+1} \rightarrow 0$. So, the power series is convergent on $(-\infty, \infty)$. ■

Theorem. For a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$, there are only the following three possibilities.

(i) The series converges only at the centre $x = a$.

(ii) The series converges for all x in $(-\infty, \infty)$.

(iii) There is a positive number R such that the series is convergent if $|x-a| < R$ and is divergent if $|x-a| > R$. In this case, when $x = a \pm R$, the series can be convergent or divergent.

The number R in (iii) is called the radius of convergence of the power series. We can let $R = 0$ in (i), and let $R = \infty$ in (iii).

The interval of convergence of the power series consists of all values of x for which the power series is convergent. Therefore, we have the following six types of intervals of convergence:

$$\{a\}, \quad (-\infty, \infty), \quad (a-R, a+R), \quad [a-R, a+R), \quad (a-R, a+R], \quad [a-R, a+R].$$

For example, for the three power series in Example 1, $R = 0, 1$ and ∞ , respectively, and the intervals of convergence are $\{0\}, [2, 4)$ and $(-\infty, \infty)$, respectively.

Example 2. For the power series

$$\sum_{n=0}^{\infty} x^n, \quad \sum_{n=1}^{\infty} \frac{1}{n} x^n, \quad \sum_{n=1}^{\infty} \frac{1}{n^2} x^n, \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n} x^n,$$

the intervals of convergence are $(-1, 1), [-1, 1), [-1, 1]$ and $(-1, 1]$, respectively. ■

Example 3. Find the the radius of convergence and the interval of convergence.

$$(1) \quad \sum_{n=1}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}.$$

Now the centre is 0. For $x \neq 0$, we have

$$\left| \frac{a_{n+1}}{a_n} \right| = 3|x| \sqrt{\frac{n+1}{n+2}} = 3|x| \sqrt{\frac{1+1/n}{1+2/n}} \rightarrow 3|x|.$$

Therefore, the power series is convergent when $3|x| < 1$, or $|x| < \frac{1}{3}$; so, $R = \frac{1}{3}$.

When $x = -\frac{1}{3}$, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1}} = \sum_{n=2}^{\infty} \frac{1}{\sqrt{n}}$ is a divergent p -series with $p = 1/2$.

When $x = \frac{1}{3}$, $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ is a convergent alternating series. Therefore, the interval of

convergence is $\left(-\frac{1}{3}, \frac{1}{3}\right]$.

$$(2) \quad \sum_{n=1}^{\infty} \frac{n(x+2)^n}{3^{n+1}}.$$

Now the centre is -2 . For $x \neq -2$, we have

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x+2|}{3} \left(1 + \frac{1}{n}\right) \rightarrow \frac{|x+2|}{3}.$$

Therefore, the power series is convergent when $\frac{|x+2|}{3} < 1$, or $|x+2| < 3$; that is,

$R = 3$ and $-5 < x < 1$.

When $x = -5$, the series $\sum_{n=1}^{\infty} (-1)^n \frac{n}{3}$ is divergent by the Test for Divergence. Similarly,

when $x = 1$, the series $\sum_{n=1}^{\infty} \frac{n}{3}$ is also divergent. Hence, the interval of convergence is $(-5, 1)$.

$$(3) \quad \sum_{n=1}^{\infty} \frac{x^n}{n(1 + \ln n)^2}.$$

Now the centre is 0. For $x \neq 0$, we have

$$\left| \frac{a_{n+1}}{a_n} \right| = |x| \frac{n}{n+1} \left(\frac{1 + \ln n}{1 + \ln(n+1)} \right)^2 \rightarrow |x| \quad (\text{as } n \rightarrow \infty).$$

Here, we can get this limit by using L'Hospital's Rule. Therefore, the power series is convergent when $|x| < 1$; that is, $R = 1$.

When $x = -1$, the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n(1 + \ln n)^2}$ is convergent.

When $x = 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n(1 + \ln n)^2}$ is convergent by the Integral Test, since

$$\int_1^{\infty} \frac{1}{x(1 + \ln x)^2} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x(1 + \ln x)^2} dx = 1 - \lim_{t \rightarrow \infty} \frac{1}{1 + \ln t} = 1 < \infty.$$

Therefore, the interval of convergence is $[-1, 1]$. ■

Section 11.9: Representing Functions as Power Series

Recall that when $-1 < x < 1$, we have $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n$.

Using this equality, we can represent some functions as power series.

Example 1. Find a power series representation for each of the following functions.

(1) $\frac{1}{1+x^2}$. We have

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} \quad (|x| < 1 \text{ since } x^2 < 1).$$

(2) $\frac{1}{2+x}$. We have

$$\frac{1}{2+x} = \frac{1}{2} \frac{1}{1-\left(-\frac{x}{2}\right)} = \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n \quad (|x| < 2, \text{ since } \left|-\frac{x}{2}\right| < 1).$$

(3) $\frac{x^3}{2+x}$. We have

$$\frac{x^3}{2+x} = x^3 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^{n+3} = \sum_{n=3}^{\infty} \frac{(-1)^{n-1}}{2^{n-2}} x^n \quad (|x| < 2).$$

(4) $\frac{x}{2x^2+1}$. We have

$$\frac{x}{2x^2+1} = x \frac{1}{1-(-2x^2)} = x \sum_{n=0}^{\infty} (-2x^2)^n = \sum_{n=0}^{\infty} (-2)^n x^{2n+1} \quad (|x| < \frac{1}{\sqrt{2}}, \text{ since } |-2x^2| < 1).$$

(5) $\frac{2x+3}{x^2+3x+2}$. We write

$$\frac{2x+3}{x^2+3x+2} = \frac{2x+3}{(x+2)(x+1)} = \frac{A}{x+2} + \frac{B}{x+1}.$$

Thus $2x+3 = A(x+1) + B(x+2)$. Taking $x = -2$, we get $A = 1$; taking $x = -1$, we get $B = 1$. Therefore, we have

$$\begin{aligned} \frac{2x+3}{x^2+3x+2} &= \frac{1}{x+2} + \frac{1}{x+1} = \frac{1}{2} \frac{1}{1-\left(\frac{-x}{2}\right)} + \frac{1}{1-(-x)} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{-x}{2}\right)^n + \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} \left(\frac{(-1)^n}{2^{n+1}} + (-1)^n\right) x^n \quad (|x| < 1). \quad \blacksquare \end{aligned}$$

The reason we are interested in power series representations is the following properties of a function that can be defined by a power series.

Theorem. Suppose the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$.

Then the function $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ is differentiable on $(a-R, a+R)$. In this case,

we have

$$(i) \quad f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1};$$

$$(ii) \quad \int f(x) dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}.$$

Now both power series in (i) and (ii) also have the radius of convergence R .

Example 2. Find a power series representation for each of the following functions.

(1) $\frac{1}{(1-x)^2}$. Now $\frac{1}{(1-x)^2} = \left(\frac{1}{1-x}\right)'$ and $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$. Thus we have

$$\frac{1}{(1-x)^2} = \left(\sum_{n=0}^{\infty} x^n\right)' = \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty} (n+1) x^n \quad (|x| < 1).$$

(2) $\ln(1-x)$. Now $(\ln(1-x))' = \frac{-1}{1-x} = -\sum_{n=0}^{\infty} x^n$. Thus we have

$$\ln(1-x) = \int \frac{-1}{1-x} dx = C - \sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1} = C - \sum_{n=1}^{\infty} \frac{1}{n} x^n \quad (|x| < 1).$$

When $x = 0$, $\ln(1-x) = 0$. Therefore, $C = 0$ and

$$\ln(1-x) = \sum_{n=1}^{\infty} \frac{(-1)}{n} x^n \quad (|x| < 1).$$

Replacing x by $-x$, we have

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n \quad (|x| < 1).$$

Taking $x = -\frac{1}{2}$, we can get $\ln 2 = \sum_{n=1}^{\infty} \frac{1}{n 2^n} = \frac{1}{2} + \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} + \frac{1}{4 \cdot 2^4} + \dots$

(3) $\tan^{-1}x$. Let $f(x) = \tan^{-1}x$. Then $f'(x) = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$. Thus

$$f(x) = \int \frac{1}{1+x^2} dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \quad (|x| < 1).$$

When $x = 0$, $f(x) = 0$. Therefore, $C = 0$ and

$$f(x) = \tan^{-1}x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \quad (|x| < 1). \quad \blacksquare$$

Example 3. Evaluate $\int \frac{1}{1+x^7} dx$ as a power series.

Now $\frac{1}{1+x^7} = \frac{1}{1-(-x^7)} = \sum_{n=0}^{\infty} (-x^7)^n = \sum_{n=0}^{\infty} (-1)^n x^{7n} \quad (|x| < 1)$. Hence, we have

$$\int \frac{1}{1+x^7} dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{7n+1} x^{7n+1} \quad (|x| < 1). \quad \blacksquare$$

Section 11.10: Taylor and Maclaurin Series

Suppose $f(x)$ has a power series representation at a :

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots \quad (|x-a| < R).$$

Then we wonder how c_n is determined by f . Now we have

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}, \quad f''(x) = \sum_{n=2}^{\infty} n(n-1) c_n (x-a)^{n-2},$$

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-(k-1)) c_n (x-a)^{n-k}, \quad \dots$$

Thus we have

$$f(a) = c_0, \quad f'(a) = c_1, \quad f''(a) = 2c_2, \quad \dots \quad f^{(k)}(a) = k! c_k.$$

Therefore, we obtain that $c_n = \frac{1}{n!} f^{(n)}(a)$ ($n = 0, 1, 2, \dots$) and

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

The above series $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$ is called the Taylor series of f at a . When $a = 0$,

it becomes $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$, called the Maclaurin series of f .

Example 1. Find the Maclaurin series of $f(x) = e^x$ and the radius of convergence.

Now for all n , $f^{(n)}(x) = e^x$ and hence $f^{(n)}(0) = 1$. Therefore, the Maclaurin series of

f is $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$, and its radius of convergence is $R = \infty$. ■

Definition. The n -th Taylor polynomial of f at a is defined by

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k,$$

and the remainder of the Taylor series is defined by $R_n(x) = f(x) - T_n(x)$.

Hence, we have

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \iff \lim_{n \rightarrow \infty} R_n(x) = 0.$$

Taylor's Inequality. If $|f^{(n+1)}(x)| \leq M$ for $|x-a| \leq d$, then the remainder $R_n(x)$ of the Taylor series of f satisfies

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1} \quad \text{for } |x-a| \leq d.$$

Note that

$$\lim_{n \rightarrow \infty} \frac{d^{n+1}}{(n+1)!} = 0.$$

Therefore, we have the following

Fact. If there exists $M > 0$ such that $|f^{(n+1)}(x)| \leq M$ for all n and all $|x-a| \leq d$, then f is equal to its Taylor series.

Example 2. Prove that $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$ (its Maclaurin series) for all x in $(-\infty, \infty)$.

Proof. Let $d > 0$ and let $M = e^d$. Then for all n and all $|x| \leq d$, we have

$$|f^{(n+1)}(x)| = e^x \leq e^d = M.$$

Therefore, by the above Fact, we have $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$ for all x in $(-\infty, \infty)$. ■

Example 3. For each of the following functions, find the Maclaurin series and prove that it represents the function for all x in $(-\infty, \infty)$.

(1) $f(x) = \sin x$.

Now $f'(x) = \cos x$, $f''(x) = -\sin x$, $f'''(x) = -\cos x$, $f^{(4)}(x) = \sin x$, $\dots$, and hence

$$f(0) = 0, \quad f'(0) = 1, \quad f''(0) = 0, \quad f'''(0) = -1, \quad f^{(4)}(0) = 0, \quad \dots$$

Therefore, the Maclaurin series is

$$x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

Since $|f^{(n+1)}(x)| \leq 1$ for all n and x , by the Fact, $f(x)$ equals its Maclaurin series for all x .

(2) $f(x) = \cos x$.

Similar to the function $\sin x$, we can find that the Maclaurin series of $f(x) = \cos x$ is

$$1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

Since $|f^{(n+1)}(x)| \leq 1$ for all n and x , by the Fact, $f(x)$ equals its Maclaurin series for all x .

(3) $f(x) = x \cos x$.

By (2), we see that $f(x) = x \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n+1}$ for all x and by the uniqueness of

the Maclaurin series, this series is the Maclaurin series of f .

(4) $f(x) = \cos^2 x$.

By (2), we see that for all x ,

$$f(x) = \frac{1}{2}(1 + \cos(2x)) = \frac{1}{2} + \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (2x)^{2n} = 1 + \sum_{n=1}^{\infty} \frac{(-4)^n}{2(2n)!} x^{2n},$$

and by the uniqueness of the Maclaurin series, this is the Maclaurin series of f . ■

Example 4. Find the Taylor series of $f(x) = \sin x$ at $\frac{\pi}{2}$.

Since

$$f(x) = \sin x = \cos\left(x - \frac{\pi}{2}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(x - \frac{\pi}{2}\right)^{2n}$$

for all x , by the uniqueness of the Taylor series, this is the Taylor series of $f(x) = \sin x$ at $\frac{\pi}{2}$. ■

We mentioned that the integral $\int e^{x^2} dx$ is “unsolvable”. Now we see that it can be expressed as a power series.

Example 5. Express $\int e^{x^2} dx$ and $\int e^{-x^2} dx$ as infinite series.

Recall that for all x , we have $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$. Thus we have

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{1}{n!} x^{2n} \quad \text{and} \quad e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n}.$$

Therefore, we have

$$\int e^{x^2} dx = C + \sum_{n=0}^{\infty} \frac{1}{(2n+1)n!} x^{2n+1}$$

and

$$\int e^{-x^2} dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)n!} x^{2n+1}. \quad \blacksquare$$